

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8"/>
5    <meta name="description" content="Data Visualisation"/>
6    <meta name="keywords" content="HTML, CSS, D3"/>
7    <meta name="author" content="Anh Khoa"/>
8    <title>LAB 5.1 D3 Updates</title>
9    <script src="https://d3js.org/d3.v7.min.js"></script>
10 </head>
11
12 <body>
13   <h1>The D3 Journey starts here...</h1>
14
15   <div style="margin-bottom: 20px;">
16     <button id="updateButton">Update Data</button>
17   </div>
18
19   <script>
20     var w = 500;
21     var h = 200;
22     var maxValue = 25;
23
24     var dataset = [14, 5, 26, 23, 9, 12, 18, 25];
25
26     var svg = d3.select("body")
27       .append("svg")
28       .attr("width", w)
29       .attr("height", h);
30
31     var xScale = d3.scaleBand()
32       .domain(d3.range(dataset.length))
33       .rangeRound([0, w])
34       .paddingInner(0.1);
35
36     var yScale = d3.scaleLinear()
37       .domain([0, maxValue])
38       .range([h, 0]);
39
40     // Draw initial bars
41     svg.selectAll("rect")
42       .data(dataset)
43       .enter()
44       .append("rect")
45       .attr("x", (d, i) => xScale(i))
46       .attr("y", d => yScale(d))
47       .attr("width", xScale.bandwidth())
48       .attr("height", d => h - yScale(d))
49       .attr("fill", "green");
50
51     // Draw initial labels
52     svg.selectAll("text")
53       .data(dataset)
54       .enter()
55       .append("text")
56       .text(d => d)
57       .attr("x", (d, i) => xScale(i) + xScale.bandwidth() / 2)
58       .attr("y", d => Math.max(yScale(d) - 5, 10))
59       .attr("text-anchor", "middle")
60       .attr("font-size", "12px")
61       .attr("fill", "black");
62
63     // Update button event
64     d3.select("#updateButton").on("click", function () {
65       dataset = d3.range(8).map(() => Math.floor(Math.random() * maxValue));
66
67       xScale.domain(d3.range(dataset.length));
68
69       // Update bars
70       svg.selectAll("rect")
71         .data(dataset)
72         .join("rect")
73         .transition()
74         .duration(500)
75         .attr("x", (d, i) => xScale(i))
76         .attr("y", d => yScale(d))
77         .attr("width", xScale.bandwidth())
78         .attr("height", d => h - yScale(d))
79         .attr("fill", "orange");
80
81       // Update labels
82       svg.selectAll("text")
83         .data(dataset)
84         .join("text")
85         .transition()
86         .duration(500)
87         .text(d => d)
88         .attr("x", (d, i) => xScale(i) + xScale.bandwidth() / 2)
89         .attr("y", d => Math.max(yScale(d) - 5, 10));
90     });
91   </script>
92
93   <footer style="color: grey; margin-top: 20px;">
94     COS30045 Data Visualisation<br>
95     Joe Bloggs
96   </footer>
97 </body>
98 </html>

```